

Project Vuna — Product Brief

*For licensed South African agri-insurers exploring an underwriting partnership for the Grow Pack product.
Draft. Not a solicitation, quote, or binding offer.*

1. Executive summary

Project Vuna is a digital platform that bundles credit, certified inputs, and parametric drought insurance into a single product — the **Grow Pack** — distributed to South African smallholder farmers through their cooperatives. We are seeking a licensed underwriting partner to issue the parametric crop policy embedded in each Grow Pack. Vuna provides distribution, onboarding, data, and automated payout execution. The underwriter provides the licence, the actuarial discipline, and the balance sheet.

2. The Grow Pack

A Grow Pack is a single bundled product issued at the start of a planting season:

Component	Typical for 2 ha maize
Certified maize seed (25 kg)	R 420
NPK fertilizer (100 kg)	R 1,150
Parametric drought insurance policy	R 85
Total cost issued on credit	R 1,655
Service fee at harvest (10%)	R 165
Total repaid at harvest	≈ R 1,820

The farmer never receives cash. Inputs are paid directly to vetted suppliers and physically delivered through the cooperative's existing logistics. Repayment is auto-deducted from the harvest sale, which is also routed through the cooperative.

The insurance policy is the smallest line item by Rand value but the most consequential by impact: without it, one drought year destroys the household.

3. The parametric insurance product

We propose a **parametric drought policy** because it is the only structure where unit economics work at smallholder scale. Indemnity insurance requires per-policy claims investigation that costs more than the premium. Parametric pays automatically against an objective measurable trigger.

Proposed structure for the pilot product:

Parameter	Default value	Notes
Crop	White maize	Most-planted SA staple; deepest historical data
Coverage period	1 Sept → 31 Mar	Eastern Cape summer planting window
Trigger	Cumulative rainfall < 70% of 30-year norm	Adjustable per region grid
Data source	SAWS + CHIRPS remote-sensing	Two independent sources; cross-validated

Parameter	Default value	Notes
Policy size	R 60–120 per hectare	Tied to seed + fertilizer cost
Payout	Up to 80% of input cost, scaled to severity	Three-tier: 30% / 60% / 80%
Geographic granularity	District (pilot) → ward (scale)	Finer-grained data ingestible

Open to negotiation: trigger threshold (60/70/80%); payout structure (linear vs stepped); whether the trigger is rainfall, temperature-stress days, or a composite; data source (SAWS / CHIRPS / ERA5 / blend).

The parametric trigger is computed **off-chain on the underwriter's side**. The underwriter computes whether the trigger has fired using whatever data they trust, then signs an attestation. Vuna's payout contract releases funds on the signature. The underwriter retains full actuarial control of the product; Vuna handles only execution and data presentation.

4. What data Vuna can provide

Data	Source	Cadence
Farmer KYC pack	Cooperative-mediated, FICA-light tier	Per farmer, at registration
Plot location (GPS centroid + ha)	Cooperative-verified	Per farmer
Crop type and variety	Farmer + supplier-confirmed	Per pack
Input delivery confirmation	Supplier signed	Per pack
Real-time weather feed for plot region	SAWS + CHIRPS via API	Daily
Harvest yield	Cooperative-attested, off-taker confirmed	End of season
Repayment status	Marketplace settlement system	Daily
Default and loss data	System-generated	End of season

All data flows are POPIA-compliant. PII is encrypted at rest, hashed before any on-chain reference, and shared with the underwriter under a Data Sharing Agreement scoped strictly to underwriting purposes.

5. Distribution and operational controls

The cooperative is the operational anchor. They are not a technology layer; they are the human layer that prevents fraud and builds farmer trust.

- bullet **Onboarding:** in-person registration through a co-op officer; KYC verified before policy issue.
- bullet **Approval:** 48-hour cooperative review on every Grow Pack request.
- bullet **Disbursement:** inputs go to suppliers, never to farmer wallets. No cash handling.
- bullet **Trigger monitoring:** dual-source weather data with alerting if sources diverge.
- bullet **Payout:** on underwriter signature; settled to farmer's cooperative-linked account.
- bullet **Repayment:** marketplace-mediated; auto-deducted from harvest sale.

Loss-control levers we expect to use in the pilot: cohort cap (max 100 farmers), regional exposure cap, individual policy cap (R 240/ha), and an underwriter-controlled kill-switch.

6. Proposed pilot

A contained, observable pilot in the 2026/27 summer planting season:

Pilot parameter	Proposed value
Region	Eastern Cape — one district, recommended Mhlontlo or Mbizana
Crop	White maize
Cohort size	50–100 farmers
Average plot	2 hectares
Total insured ha	100–200 ha
Total credit deployed	R 75,000 – R 150,000
Total premium pool (5–7% of credit)	R 5,000 – R 10,000
Coverage period	1 Sept 2026 → 31 Mar 2027
Underwriter capital at risk	Capped per pilot agreement
Reinsurance	Likely not required at this pool size
Primary success metrics	Loss ratio · repayment rate · farmer NPS · payout latency
Pilot exit / scale gate	<30% loss ratio AND >65% repayment AND zero fraud incidents >5% of pool

This pool is small enough that the underwriter can absorb the risk on its own balance sheet without external reinsurance. Scaling beyond ~500 farmers per region is where reinsurance discussions become material.

7. Partnership structures we are open to

Structure	Vuna's role	Underwriter's role	Comment
Tied intermediation	Distribution, onboarding, payout execution	Issues policy directly to farmer	Simplest. Lowest commitment.
White-label / co-branded	Distribution and operations	Issues under co-branded paper	Mid-tier complexity. Defined commercial split.
Cell captive	Holds capital in a cell, operates inside underwriter's licence	Provides licence, governance, reinsurance gateway	Highest commitment. Likely post-pilot.
MGA	Delegated underwriting authority	Carrier and balance sheet	Requires Vuna FSP licence we do not yet hold.

For the pilot, **tied intermediation or white-label** is the realistic starting point. Cell captive and MGA become discussions for the post-pilot scaling phase.

8. Regulatory positioning

- bullet **Insurance:** the parametric product is underwritten by the partner. Vuna acts as a tied intermediary or under white-label cover, in either case under FAIS oversight.
- bullet **Credit:** the credit component is fronted by an NCR-registered partner (cooperative MFI or development-finance lender). Vuna provides technology and operations.
- bullet **Crypto:** on-chain settlement runs under a partner CASP licence or under a closed pilot until Vuna's own CASP application clears.
- bullet **Data:** POPIA-compliant from day one. Information Officer registered.
- bullet **Cross-border:** none in the pilot. SARB pre-approval before any pan-African expansion.

Full regulatory analysis available on request.

9. Team

Tumo Mogame — Co-founder. [Background, qualifications, prior work — to be completed.]

Pitsi Kgaume — Co-founder. [Background, qualifications, prior work — to be completed.]

We are early-stage. We do not pretend otherwise. Our product proposal, regulatory analysis, technical architecture, and outreach materials are documented; we are happy to share any of these in full.

10. Open questions for the underwriter

- 1 What is your appetite for parametric crop products in 2026/27? Have you written parametric maize before, and at what scale?
- 2 What is the smallest cohort size and pool you would consider for a pilot?
- 3 What partnership structure (tied / white-label / cell / MGA) is easiest inside your existing licensing and approval framework?
- 4 What is the minimum data set you would need for an actuary to price this — and over what historical window?
- 5 What governance and reporting cadence would you expect during the pilot?
- 6 Are there reinsurance arrangements you already have that could absorb a smallholder parametric portfolio at scale?
- 7 What are the deal-killers — the things that make this a non-starter for [Insurer]?
- 8 Who internally would own this partnership through to a signed agreement?

Next steps

- 1 A second, longer conversation with your product / agriculture lead and (where possible) an actuary, in the next 2–3 weeks.
- 2 A jointly drafted pilot scoping document by end of June 2026, ahead of the 2026/27 summer planting registration window.
- 3 A non-binding memorandum of understanding by end of July 2026 if scoping aligns.

We are talking to a small number of South African agri-insurers in parallel and will be transparent about progress with each. We are not pursuing exclusivity at this stage on either side.

Tumo Mogame & Pitsi Kgaume — Co-founders, Project Vuna

[Email] · [Phone]